# ■ PackCheck Enterprise

## AI-Powered Packaging Artwork Compliance & Dieline Quality Assurance Platform

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|---|---|
| **Problem Statement:** Esko AIForge #2 ('Ready to Launch?') | **Target Domain:** FMCG, Food, Beverage, & Pharma Packaging |
| **Core Stack:** Python 3.11, PyMuPDF, ReportLab, LangGraph, Gemini 2.5 | **Database:** Local SQLite (Persistent Precedents & Audit Trail) |

## 1. Executive Summary & The Packaging Pre-Press Challenge

In consumer packaged goods (FMCG) and pharmaceutical manufacturing, releasing an incorrect packaging artwork to commercial press causes catastrophic downstream consequences: mandatory FDA warning letters, Class I/II food recalls, destruction of pre-printed carton inventory, and wasted printing cylinder/plate engraving fees ($5,000 to $50,000 per SKU run). QA teams must reconcile two fundamentally disconnected artifacts: **(1) The PLM Master Specification (.xlsx)** containing formulation ratios, allergens, and legal marketing claims, and **(2) The Vector Dieline Proof (.pdf)** containing cuts, creases, printer marks, bleed limits, and graphic panels. **PackCheck** automates this verification end-to-end.

## 2. End-to-End System Architecture

PackCheck implements a **hybrid deterministic + multimodal AI architecture**. Math, barcode checksums, and federal rounding are calculated deterministically in pure Python (0% calculation hallucination). Semantic claim validity, typographic hierarchy, and visual contrast are evaluated using Google Gemini 2.5 Flash Multimodal Vision and a stateful LangGraph pipeline.

| Pipeline Stage | Module | Technical Mechanics & Output |
|---|---|---|
| 1. Artwork Ingestion | artwork.py | Extracts vector text, fonts, and bounding boxes via PyMuPDF; renders 200 DPI PNG proofs. |
| 2. Guide Filtering | dieline_filter.py | Purges technical dieline marks ('Cut Line', 'Glue Flap', 'Crease') to prevent spurious copy errors. |
| 3. Spec Ingestion | contentsheet.py | Parses PLM Excel sheets (.xlsx) via openpyxl: product identity, allergens, claims, UPC, nutrition. |
| 4. Nutrition Parsing | nutrition_parser.py | Regex & structural column formatter extracting vertical FDA Nutrition Facts: calories, servings, %DV. |
| 5. Deterministic Checks | nutrition_math.py<br>consistency.py<br>claims.py | FDA 21 CFR 101.9 %DV check, Atwater caloric energy, GS1 Modulo-10 UPC, Vegan vs Honey, Net Wt dual units. |
| 6. Visual Geometry | design_language.py | Calculates PDP horizontal centering offset % (cx_text vs cx_panel) and dieline crease margin safety (<8pt). |
| 7. Multimodal Vision | vision_verify.py | Crops rendered panel at 200 DPI; queries Gemini 2.5 Flash Vision for contrast (0-100), hierarchy, defects. |
| 8. Anti-Hallucination | verdict_policy.py | Dual-Quotation Rule: LLM FAILs must quote verbatim artwork and rulebook text; unverified FAILs downgraded. |
| 9. SQLite Precedents | storage.py | Persists reviewer waivers in data/packcheck.db; retrieves past historical SKU precedents matching rule ID. |
| 10. Grounded Q&A | qa.py | LangGraph state machine: classify_intent -> retrieve_context -> generate_response -> verify_grounding. |

## 3. Regulatory Mathematics & Compliance Formulas

Packaging compliance requires mathematical certainty. PackCheck implements deterministic algorithms based on federal statutes:

### A. FDA 21 CFR § 101.9 Daily Value (%DV) Calculation

The FDA mandates that % Daily Value declarations on Nutrition Facts panels correspond strictly to Reference Daily Intakes (RDIs) in 21 CFR § 101.9(c)(9). PackCheck evaluates every nutrient with the exact formula:

**Formula:** Computed %DV = (Nutrient Amount / FDA Reference Daily Intake) * 100
*Real-world Example:* Cholesterol (45.0mg / 300.0mg RDI) * 100 = **15.0%** (Artwork prints 26.0% -> **FAIL**)
*Official RDIs:* Total Fat: 78g | Sat Fat: 20g | Cholesterol: 300mg | Sodium: 2300mg | Carbs: 275g | Fiber: 28g | Added Sugars: 50g | Potassium: 4700mg

### B. Subcomponent Constraints & Caloric Energy Bounds (Atwater Factors)

- **Fat Subcomponents:** Saturated Fat + Trans Fat ≤ Total Fat + 0.5g federal rounding tolerance.
- **Carbohydrate Subcomponents:** Dietary Fiber + Total Sugars ≤ Total Carbohydrate + 0.5g tolerance.
- **Atwater Caloric Validation:** Calories_theoretical = (4 * Carbs) + (4 * Protein) + (9 * Fat). Discrepancies >25% trigger regulatory review.

### C. GS1 Modulo-10 Barcode Check Digit Algorithm

Every retail package carrying a UPC-A barcode must feature a valid 12th check digit under GS1 General Specifications Section 7.9.1:

**Step 1:** Sum odd position digits (d1, d3, d5, d7, d9, d11) and multiply by 3.
**Step 2:** Sum even position digits (d2, d4, d6, d8, d10).
**Step 3:** Total = (3 * Sum_odd) + Sum_even.
**Step 4:** Check Digit = (10 - (Total mod 10)) mod 10. If computed != printed d12 -> **FAIL**.

### D. Allergen Integrity & Claims Conflicts

- **FALCPA / FASTER Act 9 Major Allergens:** Milk, Eggs, Peanuts, Tree Nuts, Wheat, Soy, Fish, Crustacean Shellfish, Sesame. If an allergen appears in the primary formulation (e.g. *Almonds*), placing it under 'May Contain' (cross-contact advisory) constitutes illegal advisory misuse.
- **Certified Vegan vs Animal Ingredients:** Cross-referenced deterministically against Honey, Gelatin, Whey, Casein, Carmine, Shellac. Presence of Honey on artwork bearing 'Certified Vegan' produces a blocking FAIL under ISO 23662 standards.
- **Net Quantity Dual Units (15 U.S.C. § 1453):** Mandates both metric (g / mL) and US avoirdupois (oz / fl oz / lb) declarations.

## 4. Visual Layout, Centering Geometry & Multimodal Vision

PackCheck evaluates visual layout and design language using exact vector geometry combined with multimodal AI:

| Inspection Layer | Methodology & Exact Math | Failure Criteria & Action |
|---|---|---|
| 1. Spatial Vector Geometry (PyMuPDF) | Computes panel boundaries [x0, x1] -> cx_panel = (x0+x1)/2.<br>Computes title bounding box -> cx_text = (tx0+tx1)/2.<br>Offset_pt = cx_text - cx_panel<br>Offset_% = (\|Offset_pt\| / Panel_Width) * 100 | If Offset_% > 6.0%, generates VIS-TITLE-OFF-CENTER with exact |
| 2. Dieline Crease Safety (Euclidean Clearance) | Measures minimum Euclidean distance from text block boundaries to dieline fold lines. | If distance < 8.0 pt (~3mm), flags VIS-DIELINE-CREASE-CROWDIN |
| 3. Multimodal AI Vision (Gemini 2.5 Flash) | Crops 200 DPI raster of the Front Panel (PDP) and feeds raw image bytes directly to Gemini. | Evaluates text legibility against background illustrations, color contra |

## 5. Persistent SQLite Database & Precedent Memory Engine

PackCheck maintains a local persistent SQLite database at data/packcheck.db with two tables:
• finding_dismissals: Records finding_id, title, category, rule_id, artwork_filename, text_snippet, waiver reason, reviewer name, timestamp.
• audit_log: Immutable chronological timeline tracking all compliance scans, reviewer dismissals, and restorations.
**Precedent Retrieval:** When an artwork is audited, the engine executes find_dismissal_precedents() matching rule IDs across historical records. If a previous reviewer dismissed a similar warning (e.g. *'Dual units waived for export UK SKU'*), the system surfaces that precedent on the finding card, displaying who signed off, the date, and the justification.

## 6. LangGraph Grounded Regulatory Assistant

The assistant is compiled as a stateful directed acyclic graph (DAG) using langgraph.graph.StateGraph:
• **Node 1 (classify_intent):** Identifies question intent: *explain*, *why_fail*, *fix_action*, *math_audit*, *rule_lookup*.
• **Node 2 (retrieve_context):** Gathers bounding boxes, computed values, formulas, and 21 CFR citations.
• **Node 3 (generate_response):** Executes Gemini 2.5 Flash with strict grounding rules, falling back to expert regulatory templates if offline.
• **Node 4 (verify_grounding):** Asserts factual evidence alignment, removes ungrounded speculation, and attaches provenance tags.

## 7. Benchmark Packaging Samples Repository

PackCheck includes 10 realistic FMCG packaging pairs in data/samples/ generated for manual user testing:

| Sample ID | Category | Packaging Format | Net Quantity | Barcode UPC-A |
|---|---|---|---|---|
| Sample01_OatMilk_Vanilla | Plant-Based Dairy | Gable-top carton | 32 FL OZ (1 QT) 946 mL | 0 12345 67890 5 |
| Sample02_ProteinBar_ChocPeanut | Sports Nutrition | Flow-wrap foil | 2.12 OZ (60g) | 8 50012 34567 1 |
| Sample03_OrganicKombucha_Ginger | Functional Beverage | Glass bottle label | 16 FL OZ (473 mL) | 0 41234 56789 2 |
| Sample04_DarkChocolate_SeaSalt | Confectionery | Folding carton box | 3.5 OZ (100g) | 7 93573 19283 4 |
| Sample05_GlutenFreeCookies_Almond | Bakery & Biscuits | Stand-up pouch | 6 OZ (170g) | 6 10293 84756 1 |
| Sample06_ColdBrewCoffee_Can | RTD Beverage | Aluminum can wrap | 11 FL OZ (325 mL) | 8 60001 23456 7 |
| Sample07_GreekYogurt_HoneyBerry | Cultured Dairy | Tub sleeve | 5.3 OZ (150g) | 0 71111 22233 4 |
| Sample08_ArtisanPasta_Spelt | Dry Pasta | Window carton box | 16 OZ (1 LB) 454g | 8 12345 67891 2 |
| Sample09_SparklingWater_Lime | Flavored Water | Beverage can | 12 FL OZ (355 mL) | 0 54321 98765 4 |
| Sample10_EnergyBar_MatchaBerry | Organic Snack | Foil wrapper | 1.76 OZ (50g) | 8 98765 43210 9 |

## 8. Test Verification & Architectural Roadmap

• **Test Verification:** All 22 tests in tests/ pass cleanly, verifying Atwater bounds, trans fat limits, GS1 check digits, FALCPA advisories, dieline filters, and anti-hallucination dual-quote rules. 0 lint errors across 52 codebase files.
• **Roadmap for Enterprise Deployment:**
1. *Dynamic CAD Layer Polygons:* Replace coordinate panel slicing with graph cycle traversal on structural CAD dielines.
2. *OCR Fallback Engine:* Deploy Gemini Document OCR for flattened artworks with outlined vector typography.
3. *Vector Embeddings for Precedents:* Integrate sqlite-vec for semantic similarity search over historical waivers.
4. *EU FIC Regulation 1169/2011:* Inspect font-weight tags for bold inline allergen highlighting in European packaging.